From 1D to 2D: Building the LS Tensor Product Interaction Surface

Purpose

This document gives the complete mathematical construction of the bivariate interaction term $f_{u,v}(x_u, x_v)$ using tensor products of the Lancaster–Šalkauskas (LS) natural cubic spline basis. It starts from the 1D LS basis (which you already have working) and builds up to the 2D case step by step.

No Bayesian priors, no MCMC, no code — just the construction, the design matrices, the identifiability constraints, and the penalty. The Bayesian layer comes later.

Part I: The 1D LS Basis (Recap)

Everything in Part I is already implemented in `ls_basis.R`. This recap establishes the notation that Part II will build on.

Step 1. Knots

Fix a covariate index $j \in \{1, \ldots, p\}$. Choose $M_j \geq 4$ knot locations:

$$\tau_{1j} < \tau_{2j} < \cdots < \tau_{M_j,j}$$

typically equally spaced on $[\min(X_j), \max(X_j)]$.

Define the knot spacings:

$$h_{mj} = \tau_{mj} - \tau_{m-1,j}, \qquad m = 2, \ldots, M_j$$

Step 2. Raw LS Basis Functions

The LS construction produces two families of compactly supported cubic spline basis functions evaluated at any scalar x:

- $\Phi_{mj}(x)$ for $m = 1, \ldots, M_j$ — the **value basis**
- $\Psi_{mj}(x)$ for $m = 1, \ldots, M_j$ — the **slope basis**

These satisfy the **cardinal properties**:

$$\Phi_{mj}(\tau_{\ell j}) = \delta_{m\ell}, \qquad \Phi'_{mj}(\tau_{\ell j}) = 0$$

$$\Psi_{mj}(\tau_{\ell j}) = 0, \qquad \Psi'_{mj}(\tau_{\ell j}) = \delta_{m\ell}$$

So Φ_{mj} is 1 at knot m and 0 at all other knots (and has zero slope everywhere at knots), while Ψ_{mj} is 0 at all knots but has slope 1 at knot m and slope 0 at all other knots.

Any natural cubic spline on these knots can be written as:

$$f_j(x) = \sum_{m=1}^{M_j} \theta_{mj}\, \Phi_{mj}(x) + \sum_{m=1}^{M_j} \gamma_{mj}\, \Psi_{mj}(x)$$

where $\theta_{mj} = f_j(\tau_{mj})$ are the knot ordinates (function values at knots) and $\gamma_{mj} = f_j'(\tau_{mj})$ are the knot slopes.

Step 3. Eliminating the Slopes via A and C

The representation in Step 2 has $2M_j$ parameters (M_j ordinates + M_j slopes), but a natural cubic spline on M_j knots has only M_j degrees of freedom. The natural boundary conditions $f_j''(\tau_{1j}) = 0$ and $f_j''(\tau_{M_j,j}) = 0$, together with continuity of f_j'' at all interior knots, impose a linear relationship between slopes and ordinates:

$$A_j\, \gamma_j = C_j\, \theta_j$$

where:

- $A_j \in \mathbb{R}^{M_j \times M_j}$ is a **tridiagonal** matrix (diagonal entries are all 2; off-diagonal entries involve the ω-weights derived from knot spacings)
- $C_j \in \mathbb{R}^{M_j \times M_j}$ is **sparse** (tridiagonal with a factor of 3; entries involve ω-weights and knot spacings)

Since A_j is tridiagonal and strictly diagonally dominant, it is invertible, so:

$$\gamma_j = A_j^{-1} C_j\, \theta_j$$

This eliminates the slopes entirely. The spline is now determined by the M_j knot ordinates $\theta_j = (\theta_{1j}, \ldots, \theta_{M_j,j})^\top$ alone.

Step 4. The Reduced LS Design Vector

Substitute $\gamma_j = A_j^{-1} C_j\, \theta_j$ back into the spline representation:

$$f_j(x) = \underbrace{\left[\Phi_j(x)^\top + \Psi_j(x)^\top A_j^{-1} C_j\right]}_{z_j(x)^\top} \theta_j$$

where:

- $\Phi_j(x) = \big(\Phi_{1j}(x), \ldots, \Phi_{M_j,j}(x)\big)^\top \in \mathbb{R}^{M_j}$
- $\Psi_j(x) = \big(\Psi_{1j}(x), \ldots, \Psi_{M_j,j}(x)\big)^\top \in \mathbb{R}^{M_j}$

The vector $z_j(x) \in \mathbb{R}^{M_j}$ is the **reduced LS basis vector** at the point x.

Key property (cardinality of z_j):

$$z_{m'j}(\tau_{mj}) = \delta_{m'm}$$

This follows directly from the cardinal properties of Φ and Ψ: at a knot τ_{mj}, $\Phi_{m'j}(\tau_{mj}) = \delta_{m'm}$ and $\Psi_{\ell j}(\tau_{mj}) = 0$ for all ℓ, so the $\Psi A^{-1} C$ term vanishes and only the Φ term survives.

What this means: $f_j(\tau_{mj}) = z_j(\tau_{mj})^\top \theta_j = \theta_{mj}$. Each coefficient θ_{mj} is literally the function value at knot m. This is the **knot-ordinate property** — unique to the LS basis. B-spline coefficients do not have this interpretation.

Step 5. The Design Matrix Z_j

For n observations $x_{1j}, x_{2j}, \ldots, x_{nj}$, stack the row vectors:

$$Z_j = \begin{pmatrix} z_j(x_{1j})^\top \\ z_j(x_{2j})^\top \\ \vdots \\ z_j(x_{nj})^\top \end{pmatrix} \in \mathbb{R}^{n \times M_j}$$

The vector of fitted values is $\mathbf{f}_j = Z_j\, \theta_j \in \mathbb{R}^n$.

Step 6. Identifiability: The Contrast Matrix T_j

The additive model includes an intercept μ. Since Z_j contains the constant function in its column space (because the knot ordinates can all be set to the same value), the intercept and the function f_j are not separately identified without a constraint.

We impose **sum-to-zero on the knot ordinates**:

$$\sum_{m=1}^{M_j} \theta_{mj} = 0 \qquad \Longleftrightarrow \qquad \mathbf{1}_{M_j}^\top \theta_j = 0$$

This is enforced by reparameterizing:

$$\theta_j = T_j\, \beta_j$$

where $T_j \in \mathbb{R}^{M_j \times (M_j - 1)}$ is the contrast matrix:

$$T_j = \begin{pmatrix} -1 & -1 & \cdots & -1 \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

The first row is all -1's; rows 2 through M_j form I_{M_j-1}.

Critical property:

$$\mathbf{1}_{M_j}^\top T_j = \mathbf{0}^\top$$

Proof: $\mathbf{1}^\top T_j = (-1, -1, \ldots, -1) + (1, 0, \ldots, 0) + (0, 1, \ldots, 0) + \cdots + (0, 0, \ldots, 1) = \mathbf{0}^\top$. $\square$

So for any β_j, we have $\mathbf{1}^\top \theta_j = \mathbf{1}^\top T_j \beta_j = 0$.

The **identified design matrix** is:

$$W_j = Z_j\, T_j \in \mathbb{R}^{n \times (M_j - 1)}$$

and the model becomes $\mathbf{f}_j = W_j\, \beta_j$ with $\beta_j \in \mathbb{R}^{M_j - 1}$.

Step 7. The 1D RW2 Penalty

To encourage smooth functions, we penalize the curvature of the knot ordinates. Since $\theta_{mj} = f_j(\tau_{mj})$, second differences of θ_j approximate the second derivative of f_j:

$$\Delta^2 \theta_{mj} = \theta_{m+1,j} - 2\,\theta_{mj} + \theta_{m-1,j} \;\approx\; h^2\, f_j''(\tau_{mj})$$

The RW2 penalty is:

$$\mathcal{P}_j = \sum_{m=2}^{M_j - 1} \left(\Delta^2 \theta_{mj}\right)^2 = \theta_j^\top K_j^{(\theta)}\, \theta_j$$

where $K_j^{(\theta)} = (D_2^{(j)})^\top D_2^{(j)}$ and $D_2^{(j)} \in \mathbb{R}^{(M_j - 2) \times M_j}$ is the second-difference operator:

$$D_2^{(j)} = \begin{pmatrix} 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ & & \ddots & \ddots & \ddots & \\ 0 & \cdots & 0 & 1 & -2 & 1 \end{pmatrix}$$

In terms of the identified coefficients $\beta_j = T_j^{-\ell}\theta_j$ (where $\theta_j = T_j\beta_j$):

$$\mathcal{P}_j = \beta_j^\top \underbrace{T_j^\top K_j^{(\theta)} T_j}_{K_j^{(\beta)}} \beta_j$$

The matrix $K_j^{(\beta)} \in \mathbb{R}^{(M_j-1)\times(M_j-1)}$ is the penalty expressed in identified coordinates.

Rank: $D_2^{(j)}$ has $M_j - 2$ rows and M_j columns, so $K_j^{(\theta)}$ has rank $M_j - 2$ (null space = linear functions, dimension 2). After the T_j transformation, $K_j^{(\beta)}$ has rank $(M_j - 1) - 1 = M_j - 2$ (null space dimension 1, corresponding to the single linear direction that T_j preserves).

End of Part I. Everything above is implemented and tested. Part II now extends this to two dimensions.

Part II: The 2D Tensor Product Construction

We want to build an interaction surface $f_{u,v}(x_u, x_v)$ for two covariates u and v. The idea: take the 1D LS machinery for covariate u (knots, $A_u^{-1}C_u$, basis vector z_u, contrast T_u) and the 1D LS machinery for covariate v (knots, $A_v^{-1}C_v$, basis vector z_v, contrast T_v), and combine them via tensor products.

Step 8. The 2D Basis: Products of 1D Basis Vectors

The fundamental object is the **product of two 1D reduced LS basis functions**:

$$c_{ml}(x_u, x_v) = z_{um}(x_u) \cdot z_{vl}(x_v)$$

where $z_{um}(x_u)$ is the m-th component of $z_u(x_u) \in \mathbb{R}^{M_u}$ and $z_{vl}(x_v)$ is the l-th component of $z_v(x_v) \in \mathbb{R}^{M_v}$.

There are $M_u \times M_v$ such product functions, one for each pair (m, l) with $m = 1, \ldots, M_u$ and $l = 1, \ldots, M_v$.

Why this works (Lancaster Ch. 8, Theorem 8.2.1): If the 1D basis functions $\{z_{um}\}$ satisfy the cardinality property on knots $\tau_{1u}, \ldots, \tau_{M_u,u}$ and the 1D basis functions $\{z_{vl}\}$ satisfy the cardinality property on knots $\tau_{1v}, \ldots, \tau_{M_v,v}$, then the product functions $\{c_{ml}\}$ satisfy the **2D cardinality property** on the grid of knot pairs:

$$c_{ml}(\tau_{m'u}, \tau_{l'v}) = z_{um}(\tau_{m'u}) \cdot z_{vl}(\tau_{l'v}) = \delta_{mm'} \cdot \delta_{ll'} = \begin{cases} 1 & \text{if } m = m' \text{ and } l = l' \\ 0 & \text{otherwise} \end{cases}$$

Step 9. The Interaction Surface

Define the interaction surface as a linear combination of these 2D basis functions:

$$f_{u,v}(x_u, x_v) = \sum_{m=1}^{M_u} \sum_{l=1}^{M_v} \theta_{ml} \; z_{um}(x_u) \; z_{vl}(x_v)$$

The coefficients θ_{ml} form an $M_u \times M_v$ matrix (or "grid") of surface heights.

The 2D knot-ordinate property. Evaluate $f_{u,v}$ at the grid point $(\tau_{m_0,u}, \tau_{l_0,v})$:

$$f_{u,v}(\tau_{m_0,u}, \tau_{l_0,v}) = \sum_{m} \sum_{l} \theta_{ml} \; \underbrace{z_{um}(\tau_{m_0,u})}_{\delta_{m,m_0}} \; \underbrace{z_{vl}(\tau_{l_0,v})}_{\delta_{l,l_0}} = \theta_{m_0,l_0}$$

So:

> **Each coefficient θ_{ml} is the surface height at the grid point (τ_{mu}, τ_{lv}).**

This is the direct 2D generalization of the 1D knot-ordinate property. It means the coefficient grid has an immediate physical interpretation: it IS the surface evaluated on the knot lattice.

Step 10. Vectorization

To work with the $M_u \times M_v$ coefficient grid as a single vector, we **vectorize** it. We use the convention that the v-index varies fastest (column-major / vec operator):

$$\theta_{uv} = \mathrm{vec}(\Theta) = (\theta_{11}, \theta_{12}, \ldots, \theta_{1,M_v}, \; \theta_{21}, \theta_{22}, \ldots, \theta_{2,M_v}, \; \ldots, \; \theta_{M_u,1}, \ldots, \theta_{M_u,M_v})^\top \in \mathbb{R}^{M_u M_v}$$

The position of θ_{ml} in this vector is at index $(m-1)M_v + l$.

Step 11. The 2D Basis Vector (Kronecker Product)

For a single observation with covariate values (x_u, x_v), the interaction surface value is:

$$f_{u,v}(x_u, x_v) = \sum_{m=1}^{M_u} \sum_{l=1}^{M_v} \theta_{ml} \; z_{um}(x_u) \; z_{vl}(x_v)$$

We can write this as a dot product with the vectorized θ_{uv}. The trick is to notice that the double sum equals:

$$f_{u,v}(x_u, x_v) = \big(z_u(x_u) \otimes z_v(x_v)\big)^\top \theta_{uv}$$

where $\otimes$ denotes the **Kronecker product** of two vectors.

Why? The Kronecker product $z_u(x_u) \otimes z_v(x_v) \in \mathbb{R}^{M_u M_v}$ has entries:

$$[z_u \otimes z_v]_{(m-1)M_v+l} = z_{um}(x_u) \cdot z_{vl}(x_v)$$

Taking the dot product with θ_{uv}:

$$(z_u \otimes z_v)^\top \theta_{uv} = \sum_{m=1}^{M_u} \sum_{l=1}^{M_v} z_{um}(x_u)\ z_{vl}(x_v)\ \theta_{ml}$$

which is exactly the double sum. □

Step 12. The Interaction Design Matrix Z_{uv}

For n observations, observation r has covariate values (x_{ru}, x_{rv}). Stack the row vectors:

$$Z_{uv} = \begin{pmatrix} \big(z_u(x_{1u}) \otimes z_v(x_{1v})\big)^\top \\ \big(z_u(x_{2u}) \otimes z_v(x_{2v})\big)^\top \\ \vdots \\ \big(z_u(x_{nu}) \otimes z_v(x_{nv})\big)^\top \end{pmatrix} \in \mathbb{R}^{n \times M_u M_v}$$

This is the **Khatri–Rao product** (row-wise Kronecker product) of Z_u and Z_v:

$$Z_{uv} = Z_u \odot Z_v$$

meaning: row r of Z_{uv} is the Kronecker product of row r of Z_u with row r of Z_v.

Important distinction from a full Kronecker product: $Z_u \otimes Z_v$ would be an $n^2 \times M_u M_v$ matrix (every pair of rows). The Khatri–Rao $Z_u \odot Z_v$ is $n \times M_u M_v$ — it only pairs each observation with itself, because each observation has ONE pair of covariate values (x_{ru}, x_{rv}).

The fitted interaction values are:

$$\mathbf{f}_{uv} = Z_{uv}\, \theta_{uv} \in \mathbb{R}^n$$

Step 13. Identifiability Constraints for the Interaction

In the additive model with interactions:

$$\eta_r = \mu + f_u(x_{ru}) + f_v(x_{rv}) + f_{u,v}(x_{ru}, x_{rv}) + \cdots$$

the interaction $f_{u,v}$ must be a **pure interaction** — it must not contain any component that could be absorbed by the main effects f_u, f_v, or the intercept μ. Otherwise the model is not identified

(infinitely many combinations of main effects and interaction could produce the same fitted values).

This requires the following **ANOVA-type constraints** on the coefficient grid $\{\theta_{ml}\}$:

Constraint (i) — no main effect of v hidden in the interaction:

$$\sum_{m=1}^{M_u} \theta_{ml} = 0 \qquad \text{for each } l = 1, \ldots, M_v$$

In words: every column of the coefficient grid sums to zero.

Why this constraint? If $\sum_m \theta_{ml} = c_l \neq 0$ for some l, then the interaction surface contains a component $\sum_l c_l \, z_{vl}(x_v)$, which is a function of x_v alone — this would be confounded with the main effect f_v.

Constraint (ii) — no main effect of u hidden in the interaction:

$$\sum_{l=1}^{M_v} \theta_{ml} = 0 \qquad \text{for each } m = 1, \ldots, M_u$$

In words: every row of the coefficient grid sums to zero.

Why this constraint? If $\sum_l \theta_{ml} = c_m \neq 0$ for some m, the interaction contains a component $\sum_m c_m \, z_{um}(x_u)$, confounded with f_u.

Constraint (iii) — no intercept hidden in the interaction:

$$\sum_{m=1}^{M_u} \sum_{l=1}^{M_v} \theta_{ml} = 0$$

This is automatically implied by either (i) or (ii), so constraints (i) and (ii) together are the complete set.

What this looks like on the grid. If $M_u = 4$ and $M_v = 3$, the coefficient grid is:

$$\Theta = \begin{pmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \\ \theta_{31} & \theta_{32} & \theta_{33} \\ \theta_{41} & \theta_{42} & \theta_{43} \end{pmatrix}$$

Constraint (i) says each column sums to zero: $\theta_{11} + \theta_{21} + \theta_{31} + \theta_{41} = 0$, etc. Constraint (ii) says each row sums to zero: $\theta_{11} + \theta_{12} + \theta_{13} = 0$, etc.

Step 14. Enforcing Identifiability via $T_u \otimes T_v$

Claim: Setting

$$\theta_{uv} = (T_u \otimes T_v)\, \beta_{uv}$$

where $\beta_{uv} \in \mathbb{R}^{(M_u-1)(M_v-1)}$ is the vector of free parameters, automatically enforces both constraints (i) and (ii).

Why $T_u \otimes T_v$? In the 1D case, T_j enforces $\mathbf{1}^\top \theta_j = 0$ because $\mathbf{1}^\top T_j = \mathbf{0}^\top$. In 2D, we need $\mathbf{1}^\top$ applied to every column (constraint i) and every row (constraint ii) of the grid. The Kronecker product $T_u \otimes T_v$ does exactly this — it applies T_u to the "row direction" and T_v to the "column direction" simultaneously.

Proof of Constraint (i):

Fix a column index l. We need $\sum_{m=1}^{M_u} \theta_{ml} = 0$.

The Kronecker structure means the (m, l)-th entry of the grid is:

$$\theta_{ml} = \sum_{i=1}^{M_u-1} \sum_{j=1}^{M_v-1} [T_u]_{mi}\, [T_v]_{lj}\, \beta_{ij}$$

Sum over m:

$$\sum_{m=1}^{M_u} \theta_{ml} = \sum_{i=1}^{M_u-1} \sum_{j=1}^{M_v-1} \underbrace{\left(\sum_{m=1}^{M_u} [T_u]_{mi} \right)}_{=[\mathbf{1}^\top T_u]_i \,=\, 0} [T_v]_{lj}\, \beta_{ij} = 0$$

The key step: $\sum_m [T_u]_{mi} = [\mathbf{1}^\top T_u]_i = 0$ by the critical property of T_u from Step 6. This kills the entire sum for every i, regardless of β_{ij}. $\square$

Proof of Constraint (ii):

Fix a row index m. Sum over l:

$$\sum_{l=1}^{M_v} \theta_{ml} = \sum_i \sum_j [T_u]_{mi} \underbrace{\left(\sum_{l=1}^{M_v} [T_v]_{lj} \right)}_{=[\mathbf{1}^\top T_v]_j \,=\, 0} \beta_{ij} = 0$$

Same argument, now using $\mathbf{1}^\top T_v = \mathbf{0}^\top$. $\square$

Constraint (iii) follows from either (i) or (ii). $\square$

Dimension count: θ_{uv} has $M_u M_v$ entries. β_{uv} has $(M_u - 1)(M_v - 1)$ entries. The number of constraints removed is $M_u M_v - (M_u - 1)(M_v - 1) = M_u + M_v - 1$, which equals the number of independent linear constraints in (i) and (ii) combined (M_v column constraints + M_u row constraints − 1 redundancy from constraint (iii) being implied).

Step 15. The Identified Interaction Design Matrix W_{uv}

Substituting $\theta_{uv} = (T_u \otimes T_v)\, \beta_{uv}$ into the fitted values:

$$\mathbf{f}_{uv} = Z_{uv}\, \theta_{uv} = Z_{uv}\, (T_u \otimes T_v)\, \beta_{uv} = W_{uv}\, \beta_{uv}$$

where:

$$W_{uv} = Z_{uv}\, (T_u \otimes T_v) \in \mathbb{R}^{n \times (M_u - 1)(M_v - 1)}$$

A useful shortcut. We can avoid ever forming the large Z_{uv} matrix. Notice that for row r:

$$\text{row } r \text{ of } W_{uv} = \left(z_u(x_{ru})^\top \otimes z_v(x_{rv})^\top\right)(T_u \otimes T_v)$$

Using the **mixed-product property** of Kronecker products, $(A^\top \otimes B^\top)(C \otimes D) = (A^\top C) \otimes (B^\top D)$:

$$= \left(z_u(x_{ru})^\top T_u\right) \otimes \left(z_v(x_{rv})^\top T_v\right) = w_u(x_{ru})^\top \otimes w_v(x_{rv})^\top$$

where $w_j(x) = T_j^\top z_j(x)$ is the 1D identified basis vector (the row of W_j).

Therefore:

$$\boxed{W_{uv} = W_u \odot W_v}$$

The identified interaction design matrix is simply the Khatri–Rao product of the two 1D identified design matrices. You never need to build Z_{uv} or $T_u \otimes T_v$ explicitly.

Dimensions: $W_u \in \mathbb{R}^{n \times (M_u - 1)}$, $W_v \in \mathbb{R}^{n \times (M_v - 1)}$, so $W_{uv} \in \mathbb{R}^{n \times (M_u - 1)(M_v - 1)}$.

With $M_u = M_v = 20$: W_{uv} has $19 \times 19 = 361$ columns.

Step 16. Verifying the Construction (What to Check)

Before building anything more, the following sanity checks confirm the construction is correct:

(a) Interpolation at grid points. Generate a known interaction surface $g(x_u, x_v)$ on the knot grid (τ_{mu}, τ_{lv}). Set $\theta_{ml} = g(\tau_{mu}, \tau_{lv})$. Then $Z_{uv}\, \theta_{uv}$ evaluated at the grid points should exactly reproduce g.

(b) Column sums of $T_u \otimes T_v$. Verify numerically that $(\mathbf{1}_{M_u} \otimes \mathbf{1}_{M_v})^\top (T_u \otimes T_v) = \mathbf{0}^\top$. This is equivalent to $(\mathbf{1}^\top T_u) \otimes (\mathbf{1}^\top T_v) = \mathbf{0}^\top \otimes \mathbf{0}^\top = \mathbf{0}^\top$.

(c) W_{uv} via two routes. Compute W_{uv} both ways — as $Z_{uv}(T_u \otimes T_v)$ and as the Khatri-Rao $W_u \odot W_v$ — and verify they agree to machine precision.

Part III: The 2D Penalty

Step 17. Motivation: Penalizing Surface Curvature

In 1D, the RW2 penalty penalizes the squared second differences of knot ordinates, which approximates the integrated squared second derivative $\int [f''(x)]^2 dx$. In 2D, we want to penalize the curvature of the interaction surface along both axes.

Since $\theta_{ml} = f_{u,v}(\tau_{mu}, \tau_{lv})$ are surface heights at the knot grid, second differences of the grid approximate second partial derivatives of the surface.

Step 18. Second Differences on the Grid

Along the u-direction (varying m, fixed l):

$$\Delta_u^2 \, \theta_{ml} = \theta_{m+1,l} - 2\,\theta_{ml} + \theta_{m-1,l} \;\approx\; h_u^2 \left. \frac{\partial^2 f_{u,v}}{\partial x_u^2} \right|_{(\tau_{mu}, \tau_{lv})}$$

Along the v-direction (varying l, fixed m):

$$\Delta_v^2 \, \theta_{ml} = \theta_{m,l+1} - 2\,\theta_{ml} + \theta_{m,l-1} \;\approx\; h_v^2 \left. \frac{\partial^2 f_{u,v}}{\partial x_v^2} \right|_{(\tau_{mu}, \tau_{lv})}$$

Step 19. The Total Penalty on θ_{uv}

Sum the squared second differences over all valid grid points in both directions:

$$\mathcal{P}(\theta_{uv}) = \underbrace{\sum_{l=1}^{M_v} \sum_{m=2}^{M_u-1} \left(\Delta_u^2 \, \theta_{ml}\right)^2}_{\text{curvature along } u} + \underbrace{\sum_{m=1}^{M_u} \sum_{l=2}^{M_v-1} \left(\Delta_v^2 \, \theta_{ml}\right)^2}_{\text{curvature along } v}$$

This is an **anisotropic** penalty — it penalizes curvature along each axis independently but does not directly penalize the cross-partial $\partial^2 f / \partial x_u \partial x_v$. (The thin-plate spline penalizes all three second-order partials jointly. The separable Kronecker penalty is standard in the P-spline and Bayesian additive model literature, following Lang & Brezger 2004.)

Step 20. Matrix Form of the 2D Penalty

Let $D_2^{(u)} \in \mathbb{R}^{(M_u-2)\times M_u}$ and $D_2^{(v)} \in \mathbb{R}^{(M_v-2)\times M_v}$ be the 1D second-difference matrices (same as in Step 7).

The 1D penalty matrices on the full (unidentified) coefficients are:

$$K_u^{(\theta)} = (D_2^{(u)})^\top D_2^{(u)} \in \mathbb{R}^{M_u\times M_u}, \qquad K_v^{(\theta)} = (D_2^{(v)})^\top D_2^{(v)} \in \mathbb{R}^{M_v\times M_v}$$

The 2D penalty matrix on the vectorized θ_{uv} is the **Kronecker sum**:

$$\boxed{K_{uv}^{(\theta)} = K_u^{(\theta)} \otimes I_{M_v} \;+\; I_{M_u} \otimes K_v^{(\theta)}}$$

Why this is the correct matrix. Let's verify each term:

- $K_u^{(\theta)} \otimes I_{M_v}$ acts on θ_{uv}: for each fixed column l of the grid, it applies the u-direction second-difference penalty to the column $(\theta_{1l}, \theta_{2l}, \ldots, \theta_{M_u,l})^\top$. The I_{M_v} factor means each column is treated independently. This produces the first sum in $\mathcal{P}$.
- $I_{M_u} \otimes K_v^{(\theta)}$ acts on θ_{uv}: for each fixed row m of the grid, it applies the v-direction second-difference penalty to the row $(\theta_{m1}, \theta_{m2}, \ldots, \theta_{m,M_v})^\top$. The I_{M_u} factor means each row is treated independently. This produces the second sum in $\mathcal{P}$.

Adding them gives the total penalty $\mathcal{P} = \theta_{uv}^\top K_{uv}^{(\theta)}\, \theta_{uv}$.

Step 21. Transforming the Penalty to Identified Coefficients

Recall $\theta_{uv} = (T_u \otimes T_v)\, \beta_{uv}$. Substitute:

$$\mathcal{P} = \theta_{uv}^\top K_{uv}^{(\theta)}\, \theta_{uv} = \beta_{uv}^\top \underbrace{(T_u \otimes T_v)^\top K_{uv}^{(\theta)}\, (T_u \otimes T_v)}_{K_{uv}^{(\beta)}} \beta_{uv}$$

Now expand $K_{uv}^{(\beta)}$ using the mixed-product property $(A \otimes B)(C \otimes D) = AC \otimes BD$:

First term:

$$(T_u \otimes T_v)^\top (K_u^{(\theta)} \otimes I_{M_v})(T_u \otimes T_v) = (T_u^\top K_u^{(\theta)} T_u) \otimes (T_v^\top I_{M_v} T_v) = K_u^{(\beta)} \otimes G_v$$

Second term:

$$(T_u \otimes T_v)^\top (I_{M_u} \otimes K_v^{(\theta)})(T_u \otimes T_v) = (T_u^\top I_{M_u} T_u) \otimes (T_v^\top K_v^{(\theta)} T_v) = G_u \otimes K_v^{(\beta)}$$

where we define:

$$K_j^{(\beta)} = T_j^\top K_j^{(\theta)} T_j \qquad \text{(1D identified penalty, already computed)}$$

$$G_j = T_j^\top T_j \qquad \text{(Gram matrix of the contrast)}$$

Result:

$$\boxed{K_{uv}^{(\beta)} = K_u^{(\beta)} \otimes G_v \;+\; G_u \otimes K_v^{(\beta)}}$$

Step 22. The Gram Matrix $G_j = T_j^\top T_j$

With the specific T_j from Step 6, where the first row is all -1's and rows $2, \ldots, M_j$ form I_{M_j-1}:

$$[G_j]_{ab} = \sum_{m=1}^{M_j} [T_j]_{ma}\, [T_j]_{mb}$$

For $a = b$: the sum is $(-1)(-1) + \delta_{a,\text{row}} = 1 + 1 = 2$ (row 1 contributes $(-1)^2 = 1$, and row $a + 1$ contributes $1^2 = 1$).

For $a \neq b$: the sum is $(-1)(-1) + 0 = 1$ (only row 1 contributes a nonzero term to both columns).

So:

$$G_j = I_{M_j-1} + \mathbf{1}\mathbf{1}^\top \in \mathbb{R}^{(M_j-1)\times(M_j-1)}$$

This is a known, fixed matrix — identity plus the all-ones matrix. It does not depend on data or knots.

Note: G_j is NOT the identity. This means $K_{uv}^{(\beta)} \neq K_u^{(\beta)} \otimes I + I \otimes K_v^{(\beta)}$. The Gram matrices introduce coupling that reflects the sum-to-zero constraint structure. If you had used an orthogonal contrast matrix instead of T_j, then $G_j = I$ and the penalty would simplify to the plain Kronecker sum. With our specific T_j, the G factors are easy to compute and store, so this is not a problem — just something to be aware of.

Step 23. Rank of $K_{uv}^{(\beta)}$

This determines the number of effectively penalized directions versus the unpenalized "null space" directions. You need the rank to correctly specify the prior distribution later (for the Bayesian layer).

1D rank recap: $K_j^{(\beta)}$ has dimension $(M_j - 1) \times (M_j - 1)$ and rank $M_j - 2$. The null space has dimension 1.

2D rank: $K_{uv}^{(\beta)} = K_u^{(\beta)} \otimes G_v + G_u \otimes K_v^{(\beta)}$ has dimension $(M_u - 1)(M_v - 1) \times (M_u - 1)(M_v - 1)$.

Because G_u and G_v are not identity matrices, the rank of this Kronecker sum is not simply computed from the ranks of the individual terms. The exact rank depends on the interaction of the null spaces through the G matrices.

Practical approach: Compute the rank numerically for your specific M values. For the typical case $M_u = M_v = 20$, the matrix is 361×361 and you can compute `qr(K_uv_beta)$rank` directly in R. The expected rank is $(M_u - 1)(M_v - 1) - 1 = 360$, with nullity 1.

Part IV: Assembling the Full Model

Step 24. The Model Equation

$$Y(s_r) = \mu + \sum_{j=1}^{p} f_j\big(X_j(s_r)\big) + \sum_{u<v} f_{u,v}\big(X_u(s_r), X_v(s_r)\big) + b(s_r) + \varepsilon_r$$

for $r = 1, \ldots, n$, where:

- μ is the intercept
- $f_j(X_j)$ are the main effects (from the 1D LS construction)
- $f_{u,v}(X_u, X_v)$ are the pairwise interaction surfaces (from the 2D tensor product construction)
- $b(s_r)$ is the spatial random effect at location s_r
- ε_r is the iid noise

Step 25. Matrix Form

$$\mathbf{y} = \mu\, \mathbf{1}_n + \sum_{j=1}^{p} W_j\, \beta_j + \sum_{u<v} W_{uv}\, \beta_{uv} + \mathbf{b} + \boldsymbol{\varepsilon}$$

where:

- $W_j \in \mathbb{R}^{n \times (M_j - 1)}$ — main effect design matrices (existing, from Steps 5–6)
- $W_{uv} \in \mathbb{R}^{n \times (M_u - 1)(M_v - 1)}$ — interaction design matrices (new, from Step 15)
- $\mathbf{b} \sim N(\mathbf{0}, \sigma^2 R(\rho, \nu))$ — spatial random effect (existing, unchanged)
- $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \tau^2 I_n)$ — nugget (existing, unchanged)

Step 26. The Combined Design Matrix

Define the full fixed-effect parameter vector:

$$\boldsymbol{\eta} = \left(\mu,\ \beta_1^\top,\ \ldots,\ \beta_p^\top,\ \beta_{12}^\top,\ \beta_{13}^\top,\ \ldots,\ \beta_{p-1,p}^\top\right)^\top$$

and the combined design matrix:

$$H = \left[\mathbf{1}_n \mid W_1 \mid \cdots \mid W_p \mid W_{12} \mid W_{13} \mid \cdots \mid W_{p-1,p}\right]$$

The model becomes:

$$\mathbf{y} = H\,\boldsymbol{\eta} + \mathbf{b} + \boldsymbol{\varepsilon}$$

This has exactly the same structure as your existing model — H just has more columns. The collapsed Gibbs sampler that marginalizes out $\mathbf{b}$ works identically; only the dimension of $\boldsymbol{\eta}$ and the block structure of the prior precision matrix change.

Step 27. Dimension Summary

Component	Coefficients per block	Number of blocks	Total
Intercept μ	1	1	1
Main effects β_j	$M_j - 1$	p	$p(M-1)$ if all $M_j = M$
Interactions β_{uv}	$(M_u - 1)(M_v - 1)$	$\binom{p}{2}$	$\binom{p}{2}(M-1)^2$ if all $M_j = M$

Example with $p = 2, M = 20$:

- Intercept: 1
- Main effects: $2 \times 19 = 38$
- Interactions: $1 \times 19^2 = 361$
- **Total: 400** columns in H, with $n = 1000$ rows. Well-conditioned.

Example with $p = 6, M = 20$:

- Intercept: 1
- Main effects: $6 \times 19 = 114$
- Interactions: $15 \times 361 = 5{,}415$
- **Total: 5,530** columns in H, with $n = 1000$ rows. Severely underdetermined — regularization from the penalties is essential. Start with $p = 2$ first.

Notation Reference

Symbol	Meaning	Dimension
M_j	Number of knots for covariate j	scalar
τ_{mj}	m-th knot for covariate j	scalar
$z_j(x)$	Reduced LS basis vector (1D)	$M_j \times 1$
Z_j	1D design matrix (unreduced)	$n \times M_j$
T_j	Sum-to-zero contrast	$M_j \times (M_j - 1)$
$G_j = T_j^\top T_j$	Gram matrix of contrast	$(M_j - 1) \times (M_j - 1)$
$W_j = Z_j T_j$	1D identified design matrix	$n \times (M_j - 1)$
β_j	1D identified coefficients	$(M_j - 1) \times 1$
$K_j^{(\theta)}$	RW2 penalty on θ_j	$M_j \times M_j$
$K_j^{(\beta)} = T_j^\top K_j^{(\theta)} T_j$	RW2 penalty on β_j	$(M_j - 1) \times (M_j - 1)$
$Z_{uv} = Z_u \odot Z_v$	2D interaction design (Khatri–Rao)	$n \times M_u M_v$
$W_{uv} = W_u \odot W_v$	2D identified interaction design	$n \times (M_u - 1)(M_v - 1)$
β_{uv}	Identified interaction coefficients	$(M_u - 1)(M_v - 1) \times 1$
$K_{uv}^{(\theta)}$	2D penalty on θ_{uv} (Kronecker sum)	$M_u M_v \times M_u M_v$
$K_{uv}^{(\beta)}$	2D penalty on β_{uv}	$(M_u - 1)(M_v - 1) \times (M_u - 1)(M_v - 1)$
$\odot$	Khatri–Rao (row-wise Kronecker) product	—
$\otimes$	Kronecker product	—